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Studierichting: Informatica

Oefeningenreeks 4A nr. 43.9

$$\int \frac{x^2(x+2)}{x^4-2x^2+1} dx$$

$$x^4-2x^2+1 = (x^2-1)^2 = (x-1)^2(x+1)^2$$

$$\frac{x^2(x+2)}{(x-1)^2(x+1)^2} = \frac{A}{(x-1)^2} + \frac{B}{x-1} + \frac{C}{(x+1)^2} + \frac{D}{x+1}$$

$$A = \left. \frac{x^2(x+2)}{(x+1)^2} \right|_{x=1} = \frac{3}{4}$$

$$B = \left. \frac{\frac{x^2(x+2)}{(x+1)^2} - \frac{3}{4}}{x-1} \right|_{x=1} = \left. \frac{4x^3+5x^2-6x-3}{4(x+1)^2(x-1)} \right|_{x=1} = \left. \frac{(x-1)(4x^2+9x+3)}{4(x+1)^2(x-1)} \right|_{x=1}$$

$$B = 1$$

$$C = \left. \frac{x^2(x+2)}{(x-1)^2} \right|_{x=-1} = \frac{1}{4}$$

$$D = \left. \frac{\frac{x^2(x+2)}{(x-1)^2} - \frac{1}{4}}{x+1} \right|_{x=-1} = \left. \frac{4x^2(x+2) - (x-1)^2}{4(x-1)^2(x+1)} \right|_{x=-1}$$

$$= \left. \frac{(x+1)(4x^2+3x-1)}{4(x-1)^2(x+1)} \right|_{x=-1}$$

$$= \left. \frac{4x^2+3x-1}{4(x-1)^2} \right|_{x=-1}$$

$$D = 0$$

$$\begin{aligned} &\Rightarrow \int \left(\frac{\frac{3}{4}}{(x-1)^2} + \frac{1}{x-1} + \frac{\frac{1}{4}}{(x+1)^2} \right) dx \\ &= \frac{3}{4} \int \frac{dx}{(x-1)^2} + \int \frac{dx}{x-1} + \frac{1}{4} \int \frac{dx}{(x+1)^2} \\ &= \ln|x-1| - \frac{3}{4(x-1)} - \frac{1}{4(x+1)} + k \end{aligned}$$

References